

ASHUTOSH AMALE

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Professional Summary

Aspiring Data Scientist & AI Engineer · CGPA 8.96 · SAKEC Mumbai

- Built end-to-end ML pipelines from raw data ingestion and feature engineering to model training, evaluation, and live FastAPI inference on health and retail datasets.
- Developed 3 production applications (React · Node.js · FastAPI · PostgreSQL) independently before Year 2, giving full-stack context for deploying AI systems.
- Presented an autonomous embedded AI project at IIT Bombay; seeking Data Science, Data Engineering, or Applied AI roles owning raw data to deployed models.

Skills

ML & AI:	Scikit-learn · Pandas · NumPy · Matplotlib · feature engineering · model deployment · classification · regression
Data Engineering:	PostgreSQL · MongoDB · SQLite · REST APIs · ETL pipelines · data cleaning · Jupyter Notebook
Languages:	Python · SQL · JavaScript · C
Backend & Deployment:	FastAPI · Node.js · Vercel · Git/GitHub · Postman
Frontend/Concepts:	React.js · HTML5 · CSS3 · Agile/Scrum · CI/CD · OOP · A/B testing · statistical modeling

Technical Projects

Heart Attack Prediction System – ML Research & Deployment | Python, Scikit-learn, Pandas, FastAPI, Vercel

- Designed a complete ML pipeline — ingestion, cleaning, feature engineering, benchmarking (Logistic Regression, Random Forest, SVM), and tuning — achieving 89% classification accuracy.
- Authored a research paper on dataset preparation, feature selection, model comparison, and accuracy analysis.
- Productionized the model through a FastAPI inference endpoint on Vercel: heart-attack-pi.vercel.app

RetailPro POS – AI-Powered Retail Intelligence System | Python, Scikit-learn, React, Node.js, FastAPI, PostgreSQL

- Built an AI-powered demand forecasting module using Scikit-learn on 12+ months of sales and seasonal data, reducing estimated manual inventory review effort by 60%.
- Engineered the data backend as sole developer: PostgreSQL schema, ETL-style sales ingestion, and REST APIs for the ML module and dashboard.
- Deployed on Vercel for billing, inventory monitoring, and AI-driven stock alerts: infotact-project1.vercel.app

Autonomous Maze-Solving Robot | Arduino, Embedded C, QTR-8 Sensor Array | SAKEC Fab Lab

- Selected to present at IIT Bombay's technical fest, one of India's high-visibility student engineering showcases.
- Implemented real-time sensor fusion and decision logic across an 8-channel IR array with motor driver PWM control.

Experience

Infotact Solutions, Mumbai

Mar 2024 – Jun 2024

Web Development Intern

- Built React.js web features and integrated REST APIs for client-facing production workflows and real-world business use cases.
- Assisted with Docker setup for local development environments, improving consistency across development and deployment workflows.
- Solved practical client requirements through debugging, UI improvements, API testing, and end-to-end feature delivery.

Education

Shah & Anchor Kutchhi Engineering College, Mumbai

B.Tech, Computer Engineering Expected May 2026 CGPA: 8.96/10.0

R.J. College of Science, Arts & Commerce, Mumbai

HSC – Science (Physics, Chemistry, Mathematics) MHT-CET: 95.45 Percentile

Achievements & Certifications

- **IIT Bombay Technical Fest** – Selected presenter for autonomous robotics project (SAKEC Fab Lab, 2024).
- **Research Publication (Under Review)** – Co-authored paper on ML-based heart attack prediction, model benchmarking, and clinical feature analysis.
- **Independent AI Deployment** – Shipped 2 live ML-powered applications before Year 2 of B.Tech.